CReF: Cross-modal and Recurrent Fusion for Depth-conditioned Humanoid Locomotion

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Abstract*— Stable traversal over geometrically complex terrain increasingly requires exteroceptive perception, yet prior perceptive humanoid locomotion methods often remain tied to explicit geometric abstractions, either by mediating control through robot-centric 2.5D terrain representations or by shaping depth learning with auxiliary geometry-related targets. Such designs inherit the representational bias of the intermediate or supervisory target and can be restrictive for vertical structures, perforated obstacles, and complex real-world clutter. We propose CReF (Cross-modal and Recurrent Fusion), a single-stage depth-conditioned humanoid locomotion framework that learns locomotion-relevant features directly from raw forward-facing depth without explicit geometric intermediates. CReF couples proprioception and depth tokens through proprioception-queried cross-modal attention, fuses the resulting representation with a gated residual fusion block, and performs temporal integration with a Gated Recurrent Unit (GRU) regulated by a highway-style output gate for state-dependent blending of recurrent and feedforward features. To further improve terrain interaction, we introduce a terrain-aware foothold placement reward that extracts supportable foothold candidates from foot-end point-cloud samples and rewards touchdown locations that lie close to the nearest supportable candidate. Experiments in simulation and on a physical humanoid demonstrate robust traversal over diverse terrains and effective zero-shot transfer to real-world scenes containing handrails, hollow pallet assemblies, severe reflective interference, and visually cluttered outdoor surroundings. Project page: `https://cometlogic.github.io/cref/`

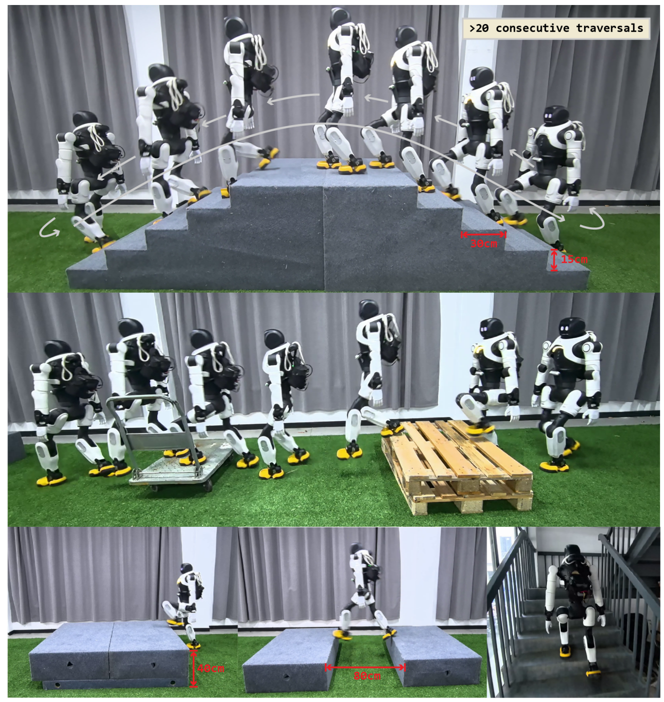

Fig. 1. Our CReF framework enables robust real-world humanoid locomotion, including more than 20 consecutive stair traversals, a 40 cm high platform, an 80 cm gap, and real-world stairs with a 20 cm rise and 26 cm tread, while generalizing robustly beyond the training terrains.

I. Introduction

Humanoid robots promise versatile mobility in human-centered environments. Deep reinforcement learning has enabled agile locomotion across diverse platforms and surface conditions, and proprioception-dominant policies can already achieve strong standing and walking performance through reactive stabilization [1]–[4]. However, perceptive locomotion becomes essential when safe traversal depends on anticipating terrain geometry before contact. This is especially critical on stairs, particularly during descent, and near gaps or abrupt height transitions, where foothold errors can lead to edge contacts or slipping [5], [6].

A widely adopted way to incorporate exteroception into locomotion is to mediate perception through an explicit 2.5D terrain representation. Early perceptive locomotion systems primarily used such representations to support foothold selection and predictive control [7], [8], while subsequent learning-based methods increasingly conditioned the controller on terrain maps or related terrain observations to extract locomotion-relevant structure from local geometry [9]–[11]. Related hybrid approaches further combined terrain perception with structured planner guidance, training learned controllers to track planner-generated foothold or motion references [12]. More recently, attention-based map encoders have extended this paradigm by enabling the policy to emphasize future steppable regions in the map [13]. This line of work is attractive because it offers a compact and interpretable interface between perception and control, supported by efficient online elevation mapping [14]. Its limitation, however, is that performance remains tightly coupled to the fidelity and expressiveness of the intermediate 2.5D representation, which is inherently restrictive for overhangs, vertical clutter, and perforated obstacles [15]–[17].

A further line of work takes depth as the exteroceptive input, but often still relies on auxiliary targets to shape what the policy learns from it [18], [19]. Because a forward-facing depth camera provides only look-ahead geometric cues and

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cannot directly observe the instantaneous underfoot region, such supervision can also complement near-foot terrain information that is otherwise unavailable from the current view. Rather than letting the locomotion objective alone organize the visual representation, these methods regularize the depth branch through terrain reconstruction [6], [20], [21], decoder-constrained latent estimation [22], or transfer from privileged teachers [23]. This strategy is attractive because it provides structured supervision that often improves optimization and transfer, yet it also anchors the learned representation to the inductive bias of the auxiliary target itself. When the target is a height map, a geometry surrogate, or a teacher already grounded in such abstractions, the encoder is rewarded primarily for preserving information that supports that target, while scene structure outside its representational scope is naturally deemphasized. In this sense, the blind spots of the intermediate are not removed, but inherited through supervision. A more direct alternative is to learn locomotion-relevant structure from depth without explicit terrain reconstruction or privileged geometric targets. Recent humanoid work has begun to pursue this route by training end-to-end policies that consume raw depth and proprioception directly, using temporal depth history to mitigate partial observability and improve traversal over challenging terrain and obstacle sequences [24]. In parallel, recent perceptive humanoid pipelines have devoted substantial effort to narrowing the sim-to-real gap of depth sensing through transfer-oriented training design and realistic sensor simulation that models stereo artifacts and calibration uncertainty [24], [25]. Although these techniques often improve transfer, they also increase training complexity and may bias robustness toward the assumed artifact distribution.

In this work, we propose CReF, a perceptive humanoid locomotion policy that directly maps onboard proprioception and forward-facing depth directly to joint position targets without explicit geometric intermediates or multi-stage supervision. CReF combines cross-modal attention, gated residual fusion, and recurrent fusion to extract locomotion-relevant depth features and regulate temporal memory according to the current locomotion state. We further introduce a terrain-aware foothold placement reward that extracts supportable foothold candidates from local foot-end point-cloud samples and shapes touchdown toward them. Unlike purely prohibitive do-not-step constraints [10], [23], it encourages landing near supportable regions and thus provides more directional supervision for contact placement. Without synthetic depth corruption during training, CReF transfers zero-shot to real hardware and remains effective in challenging real-world scenes.

This work makes the following contributions:

- A single-stage depth-conditioned humanoid locomotion framework, CReF, that enables robust terrain traversal and achieves stable performance at leading terrain-difficulty levels across all trained terrain categories.
- A terrain-aware foothold placement reward that extracts supportable foothold candidates from local foot-end point-cloud samples and guides touchdown toward supportable regions during terrain transitions, yielding substantial gains in long-duration stair traversal, especially in descent.
- Real-world zero-shot deployment of a single depth-conditioned humanoid locomotion policy across multiple terrain-transition tasks and broader unseen environments, demonstrating robust transfer beyond simulation.

II. Method

A. Overview

CReF is a single-stage depth-conditioned humanoid locomotion framework that maps onboard proprioception and forward-facing depth directly to joint position targets. It comprises an onboard actor, an asymmetric critic with privileged training signals, and a lightweight velocity estimator. The architecture is shown in Fig. 2, and training is performed with PPO [26].

1) Policy Network: At each control step t, the policy network takes the current proprioceptive observation $\mathbf{o}_t^p$ and the current forward-facing depth image $\mathbf{D}_t \in \mathbb{R}^{H\times W}$ as input:

$$\mathbf{o}_t = \left(\mathbf{o}_t^p,\ \mathbf{D}_t\right). \tag{1}$$

Following standard legged locomotion pipelines, the proprioceptive observation is defined as

$$\mathbf{o}_t^p = \Big[\boldsymbol{\omega}_t,\ \mathbf{r}_t^{\mathrm{grav}},\ \mathbf{u}_t^{\mathrm{cmd}},\ \mathbf{q}_t - \mathbf{q}_0,\ \dot{\mathbf{q}}_t,\ \mathbf{a}_{t-1}\Big], \tag{2}$$

where $\boldsymbol{\omega}_t$ is the base angular velocity, $\mathbf{r}_t^{\mathrm{grav}}$ is the gravity direction expressed in the body frame, $\mathbf{u}_t^{\mathrm{cmd}}$ is the commanded motion, $\mathbf{q}_t$ and $\dot{\mathbf{q}}_t$ are the joint positions and velocities, $\mathbf{q}_0$ is the nominal standing pose, and $\mathbf{a}_{t-1}$ is the previous action.

2) Value Network: To obtain a more accurate estimate of the state value, the value network is trained asymmetrically with privileged information available only in simulation. In addition to the proprioceptive observation $\mathbf{o}_t^p$, its input includes the ground-truth base linear velocity $\mathbf{v}_t$ and a robot-centric terrain height observation $\mathbf{m}_t$, which is defined as local elevation samples around the robot. The value-network input is

$$\mathbf{s}_t = \left[\mathbf{o}_t^p,\ \mathbf{v}_t,\ \mathbf{m}_t\right]. \tag{3}$$

3) Action space: The policy outputs an action $\mathbf{a}_t \in \mathbb{R}^{n_a}$ that parameterizes joint position targets tracked by a low-level PD controller. We adopt a residual parameterization around the nominal standing pose $\mathbf{q}_0$:

$$\mathbf{q}_t^{\mathrm{target}} = \mathbf{q}_0 + \mathbf{a}_t, \tag{4}$$

where $\mathbf{q}_0$ is the nominal standing pose and $\mathbf{q}_t^{\mathrm{target}}$ is the commanded joint position vector.

4) Reward function: Following recent analyses on robust humanoid reward design [3], [27], we adopt a low-speed-aware linear-velocity tracking term and a contact-shaping term that rewards standing while encouraging short-horizon single-foot contact during locomotion. The reward terms used in training are listed in Table I.

Here, $\mathbf{e}_v = \mathbf{u}_{xy}^{\mathrm{cmd}} - \mathbf{v}_{xy}$ is the planar velocity tracking error, and $\mathbb{I}_s = \mathbb{I}(\|\mathbf{u}_{xy}^{\mathrm{cmd}}\|_2 < v_s)$ switches to the low-speed

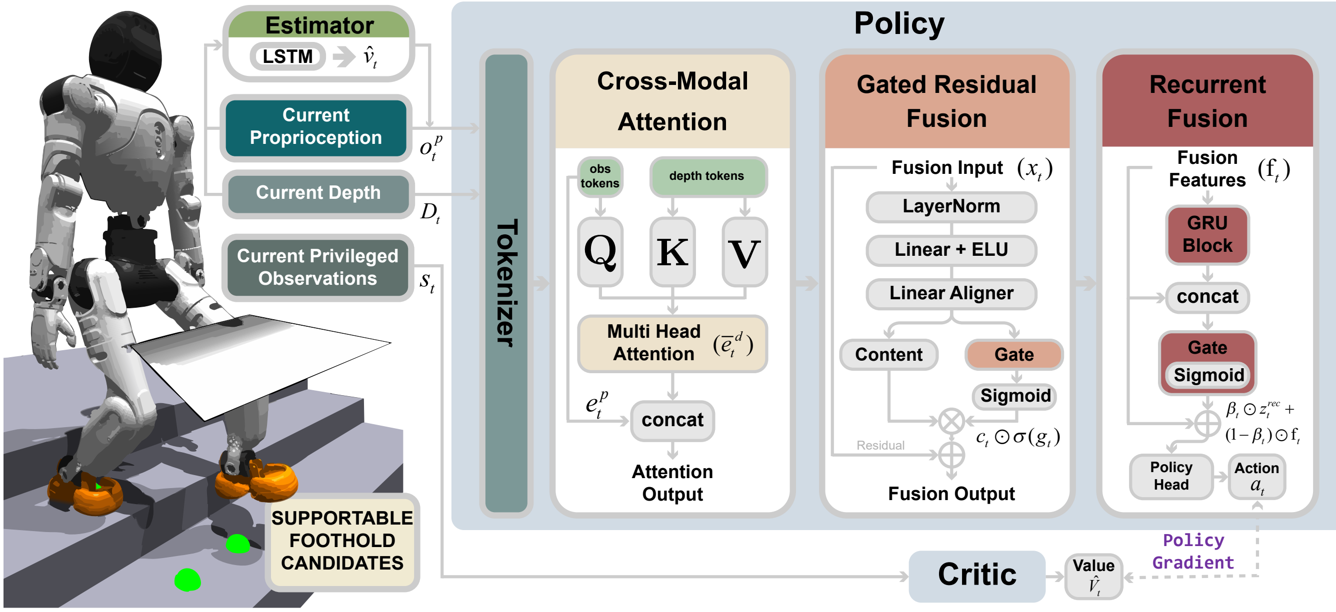

Fig. 2. Overview of CReF. The proposed single-stage depth-conditioned policy combines cross-modal attention, gated residual fusion, recurrent fusion, and a terrain-aware foothold placement reward for robust terrain locomotion.

TABLE I
REWARD TERMS USED IN TRAINING.

Term	Expression (r_i)	w_i
Lin. vel. tracking	$\exp\left(-\frac{\mathbb{I}_s \lVert \mathbf{e}_v \rVert_1 + (1-\mathbb{I}_s) \lVert \mathbf{e}_v \rVert_2^2}{\sigma_v}\right)$	2.5
Ang. vel. tracking	$\exp\left(-(\omega_z - \omega_z^{\mathrm{cmd}})^2/\sigma_\omega\right)$	1.5
Orientation	$\exp\left(-10 \lVert \mathbf{r}_{xy}^{\mathrm{grav}} \rVert_2\right)$	1.0
Feet contact	$\mathbb{I}_{\mathrm{stand}} + (1 - \mathbb{I}_{\mathrm{stand}})\, \mathbb{I}_{\mathrm{single}}^{0.2\mathrm{s}}$	1.5
Feet y distance	$\exp\left(-10 \lvert d_y^{\mathrm{feet}} - 0.27 \rvert\right)$	0.1
Foothold placement	Eq. (23)	2.0
Base height	$\exp\left(-10 \lvert (z^{\mathrm{base}} - \bar{z}^{\mathrm{supp}}) - 0.55 \rvert\right)$	0.8
Lin. vel. z	v_z^2	-1.0
Ang. vel. xy	$\lVert \boldsymbol{\omega}_{xy} \rVert^2$	-0.05
Joint velocity	$\lVert \dot{\mathbf{q}} \rVert^2$	-10^{-3}
Action rate	$\lVert \mathbf{a}_t - \mathbf{a}_{t-1} \rVert^2$	-0.01
Action smoothness	$\lVert \mathbf{a}_t - 2\mathbf{a}_{t-1} + \mathbf{a}_{t-2} \rVert^2$	-0.01
Feet air time	$\sum_f \mathbb{I}_{\mathrm{first}}^f \left(0.5 - T_{\mathrm{air}}^f\right)$	-2.5
Foot slip	$\sum_f \mathbb{I}\left(\lVert \mathbf{F}^f \rVert > 5\right) \lVert \mathbf{v}_{xyz}^f \rVert$	-0.2
Foot impact acc.	$\sum_f \left[\lvert \dot{v}_z^f \rvert - a_0 \right]_+$	-0.5
Foot impact vel.	$\sum_f \left[\lvert v_z^f \rvert - v_0 \right]_+$	-1.5
DoF position limits	$\sum_j \left(\left[q_j^{\min} - q_j\right]_+ + \left[q_j - q_j^{\max}\right]_+ \right)$	-10
DoF velocity limits	$\sum_j \left[\lvert \dot{q}_j \rvert - \dot{q}_j^{\max} \right]_+$	-0.6
Hip pos	$\sum_{j \in \mathcal{J}_{\mathrm{hip\ x/z}}} (q_j - q_j^0)^2$	-7
Ankle pos	$\sum_{j \in \mathcal{J}_{\mathrm{ank\ x}}} (q_j - q_j^0)^2$	-10
Stumble	$\mathbb{I}\left(\exists f : \lVert \mathbf{F}_{xy}^f \rVert > 5 \lvert F_z^f \rvert\right)$	-10

form when the commanded planar speed is below v_s. The standing indicator is defined as $\mathbb{I}_{\mathrm{stand}} = \mathbb{I}(\|\mathbf{u}^{\mathrm{cmd}}\|_2 < u_s)$. $\mathbb{I}_{\mathrm{single}}^{0.2\mathrm{s}}$ is equal to 1 if a single-foot contact has occurred within the last 0.2 s, and 0 otherwise. We use $\mathbf{r}_{xy}^{\mathrm{grav}}$ for the xy component of the projected gravity direction, and

$$\bar{z}^{\mathrm{supp}} = \frac{\sum_f \mathbb{I}(F_z^f > 1)\, z^f}{\sum_f \mathbb{I}(F_z^f > 1)} \tag{5}$$

for the average height of supporting feet.

In Table I, the superscript f indexes feet. Accordingly, $\mathbf{F}^f$ and $\mathbf{v}^f$ denote the contact force and velocity of foot f, while F_z^f and $\mathbf{F}_{xy}^f$ denote its normal and tangential contact-force components. $\mathbb{I}_{\mathrm{first}}^f$ indicates the first contact event of foot f after an aerial phase, and T_{air}^f is the corresponding air time. We use $[x]_+ = \max(x, 0)$ for the positive-part operator, and q_j^0 denotes the j-th component of the nominal standing pose $\mathbf{q}_0$.

B. Policy Architecture

The policy is designed to extract state-relevant terrain features from forward-facing depth observations and to adaptively control the contribution of temporal memory according to the current locomotion state. In what follows, we refer to the fusion block after cross-modal attention as gated residual fusion (GRF), and to the temporal module as recurrent fusion, which consists of GRU-based temporal integration followed by a highway output gate. The architectural components are detailed in the following subsections.

1) Depth encoding and velocity estimation: The raw depth image is normalized as $\mathbf{D}_t = \mathbf{D}_t^{\mathrm{raw}}/d_{\max} - 0.5$, where $d_{\max}$ is the preset maximum depth range. Then the normalized depth image is encoded by a lightweight CNN-based depth tokenizer $\mathcal{T}_\theta(\cdot)$ into a set of local depth tokens,

$$\mathbf{Z}_t = \mathcal{T}_\theta(\mathbf{D}_t) \in \mathbb{R}^{N \times d}, \tag{6}$$

where N is the number of tokens and d is the token dimension. These depth tokens are used by the policy for subsequent cross-modal attention.

To facilitate velocity-command tracking, we jointly train an auxiliary base-velocity estimator,

$$\hat{\mathbf{v}}_t = \mathrm{LSTM}([\mathbf{o}_t^p; \mathcal{C}_\psi(\mathbf{D}_t)]), \tag{7}$$

where $\mathcal{C}_\psi(\cdot)$ is a lightweight depth compression module. The estimator is supervised by the simulator ground-truth base linear velocity using an ℓ_2 loss.

2) Cross-modal attention for proprioception-conditioned depth feature extraction: The proprioceptive tokenizer takes the concatenation of proprioception and the estimated base linear velocity as input:

$$\mathbf{e}_t^p = \mathcal{P}_\phi([\mathbf{o}_t^p; \hat{\mathbf{v}}_t]), \qquad \mathbf{E}_t^d = \mathrm{LN}(\mathbf{Z}_t), \tag{8}$$

where $\mathcal{P}_\phi(\cdot)$ denotes the proprioceptive tokenizer and $\mathbf{E}_t^d$ is the normalized depth-token set.

Cross-modal attention then uses the proprioceptive token as the query and the depth tokens as the keys and values:

$$\mathbf{Q}_t = \mathrm{LN}(\mathbf{e}_t^p)\mathbf{W}_q, \qquad \mathbf{K}_t = \mathbf{E}_t^d\mathbf{W}_k, \qquad \mathbf{V}_t = \mathbf{E}_t^d\mathbf{W}_v, \tag{9}$$

where $\mathbf{Q}_t \in \mathbb{R}^{1\times d}$ and $\mathbf{K}_t, \mathbf{V}_t \in \mathbb{R}^{N\times d}$. Multi-head attention (MHA) then aggregates locomotion-relevant information from the depth tokens conditioned on the current proprioceptive state. The resulting attention output is

$$\bar{\mathbf{e}}_t^d = \mathrm{MHA}\big(\mathbf{Q}_t, \mathbf{K}_t, \mathbf{V}_t\big), \tag{10}$$

which is concatenated with the proprioceptive token to form the fusion input,

$$\mathbf{x}_t = [\mathbf{e}_t^p; \bar{\mathbf{e}}_t^d]. \tag{11}$$

3) Gated residual fusion and recurrent fusion: Given the fusion input $\mathbf{x}_t$, the GRF first computes

$$\tilde{\mathbf{x}}_t = \phi\big(\mathbf{W}_1\,\mathrm{LN}(\mathbf{x}_t) + \mathbf{b}_1\big), \tag{12}$$

where $\phi(\cdot)$ is ELU. This projection mixes the proprioceptive and depth-conditioned features in a shared latent space while keeping the block lightweight.

A second linear layer then produces a content branch and a gate branch,

$$\begin{bmatrix}\mathbf{c}_t\\ \mathbf{g}_t\end{bmatrix} = \mathbf{W}_2\,\tilde{\mathbf{x}}_t + \mathbf{b}_2, \qquad \mathbf{c}_t, \mathbf{g}_t \in \mathbb{R}^{2d}, \tag{13}$$

and the fusion output is

$$\mathbf{f}_t = \mathbf{x}_t + \mathbf{c}_t \odot \sigma(\mathbf{g}_t). \tag{14}$$

Here, $\mathbf{c}_t$ provides a candidate residual update, while $\sigma(\mathbf{g}_t)$ adaptively controls its channel-wise contribution. This design allows the policy to emphasize informative multimodal corrections while preserving a direct residual path for stable optimization.

Within recurrent fusion, $\mathbf{f}_t$ is first processed by a GRU for temporal integration,

$$\mathbf{h}_t = \mathrm{GRU}(\mathbf{f}_t, \mathbf{h}_{t-1}), \qquad \mathbf{z}_t^{\mathrm{rec}} = \mathbf{W}_h\mathbf{h}_t, \tag{15}$$

where $\mathbf{z}_t^{\mathrm{rec}} \in \mathbb{R}^{2d}$ is the recurrent feature. The GRU aggregates short-horizon temporal context, which is important when a single depth frame is insufficient to disambiguate terrain structure or contact timing.

The recurrent feature is then combined with the current fused feature through a highway output gate,

$$\beta_t = \sigma\big(\mathbf{W}_\beta[\mathbf{z}_t^{\mathrm{rec}}; \mathbf{f}_t] + \mathbf{b}_\beta\big), \qquad \mathbf{y}_t = \beta_t \odot \mathbf{z}_t^{\mathrm{rec}} + (1-\beta_t)\odot \mathbf{f}_t. \tag{16}$$

This gate enables state-dependent fusion between instantaneous multimodal evidence and recurrent memory, so that the policy can rely more on temporal features in ambiguous or risky phases while retaining stronger feedforward responses when current observations are already sufficient. The final action is produced by an MLP head from $\mathbf{y}_t$.

C. Terrain-Aware Foothold Placement Reward

We introduce a terrain-aware foothold placement reward to shape foot placement at touchdown. For each foot, a short buffer of point-cloud samples is maintained in the local foot frame. We then apply command-conditioned forward gating by filtering out near-foot points, where the minimum retained forward distance is set from the commanded step extent and taken here as the distance covered in 0.5 s under the current forward command. The remaining points are partitioned into overlapping candidate windows of size $24\,cm \times 10\,cm$ with a stride of $4\,cm$.

Let $\mathcal{P}_{t,k}^f = \{\mathbf{p}_{t,k,i}^f\}_{i=1}^{n_{t,k}^f}$ denote the k-th candidate window for foot f at time t, where $n_{t,k}^f$ is the number of points in that window. For each window, we compute the mean and covariance,

$$\begin{aligned}\boldsymbol{\mu}_{t,k}^f &= \frac{1}{n_{t,k}^f}\sum_{i=1}^{n_{t,k}^f}\mathbf{p}_{t,k,i}^f,\\ \boldsymbol{\Sigma}_{t,k}^f &= \frac{1}{\max(n_{t,k}^f-1,1)}\\ &\quad \sum_{i=1}^{n_{t,k}^f}\big(\mathbf{p}_{t,k,i}^f-\boldsymbol{\mu}_{t,k}^f\big)\big(\mathbf{p}_{t,k,i}^f-\boldsymbol{\mu}_{t,k}^f\big)^\top.\end{aligned} \tag{17}$$

We then perform eigen-decomposition,

$$\Sigma_{t,k}^f v_{t,k,j}^f = \lambda_{t,k,j}^f v_{t,k,j}^f, \qquad \lambda_{t,k,1}^f \le \lambda_{t,k,2}^f \le \lambda_{t,k,3}^f. \tag{18}$$

and define the roughness as

$$\rho_{t,k}^f = \sqrt{\max(\lambda_{t,k,1}^f, 0)}. \tag{19}$$

A window is accepted as a foothold candidate if it is sufficiently planar, approximately horizontal, and not recessed:

$$\rho_{t,k}^f < r_{\mathrm{th}}, \qquad |v_{t,k,1,z}^f| > \eta_{\mathrm{th}}, \qquad \mu_{t,k,z}^f > h_{\min}, \tag{20}$$

For each accepted window, the candidate foothold is set to its mean,

$$\mathbf{p}_{t,k}^{f,\star} = \boldsymbol{\mu}_{t,k}^f. \tag{21}$$

To avoid chattering, the candidate set is refreshed only at liftoff and then held fixed until the next touchdown. At touchdown, we compute the nearest planar distance in the x-z plane between the realized contact position and the candidate set,

$$d_{xz}^f = \min_k \left\| \begin{bmatrix} p_{x,t}^f \\ p_{z,t}^f \end{bmatrix} - \begin{bmatrix} p_{x,t,k}^{f,\star} \\ p_{z,t,k}^{f,\star} \end{bmatrix} \right\|_2, \tag{22}$$

and define the reward as

$$r_{\mathrm{fh}} = \sum_{f\in\mathcal{F}} I_{\mathrm{td}}^f \exp\left(-d_{xz}^f/s_{xz}\right), \tag{23}$$

where I_{td}^{f} indicates a touchdown event and s_{xz} is a tolerance scale. Although computed at touchdown, this term provides an anticipatory shaping objective by encouraging the swing foot to approach locally supportable regions before contact.

D. Training Details

CReF is trained in Isaac Gym with 4096 parallel environments at a control frequency of 50 Hz. The depth stream is rendered at 64×48 resolution and updated at 20 Hz. Training is performed on a single RTX 4090 for about 30 hours, corresponding to 20,000 iterations.

To support large-batch training, depth rendering is implemented with NVIDIA Warp [28]. Robot links are approximated as capsules, and self-occlusion is computed on the GPU via first-hit ray-capsule intersection. No synthetic depth corruption is injected during training.

III. Experiments

We evaluate CReF in simulation and on hardware. The simulation experiments examine three questions: (i) whether the proposed framework improves terrain traversal performance relative to ablated variants and a perceptive locomotion baseline, (ii) whether the foothold placement reward improves stair contact placement, and (iii) how the highway gate recruits recurrent features under different traversal conditions. We then assess zero-shot real-world transfer and out-of-distribution robustness on a physical humanoid robot.

A. Experimental Setup

Our deployment platform is an AGIBOT X2 Ultra humanoid [29]. The official specification reports a height of approximately 1.31 m, a weight of approximately 39 kg, and an official peak joint torque of 120 N·m. For reference, the Humanoid Parkour Learning (HPL) [23] baseline is deployed on the full-size Unitree H1 [30], which is approximately 1.8 m tall, weighs about 47 kg, and reports a knee torque of about 360 N·m. Due to hardware differences, HPL serves only as a terrain difficulty reference; in simulation, we re-implement it on AGIBOT X2 Ultra for controlled comparison.

For onboard sensing and inference, the robot uses a forward-facing Intel RealSense D435i mounted with a downward pitch of 50° relative to the horizontal plane, together with an NVIDIA Jetson AGX Orin for onboard policy inference. The policy is transferred zero-shot from simulation, without task-specific fine-tuning. Using the hardware timestamps provided by the RealSense, we fit a linear regression to map camera time to system time and select the depth frame with an effective latency of 20 ms.

We compare the following policy variants, all trained from scratch under matched reward, optimization, and terrain settings:

- **Full CReF**: cross-modal attention + gated residual fusion (GRF) + recurrent fusion.
- **w/o Cross-Attn**: removes cross-modal attention while preserving the remaining architecture.

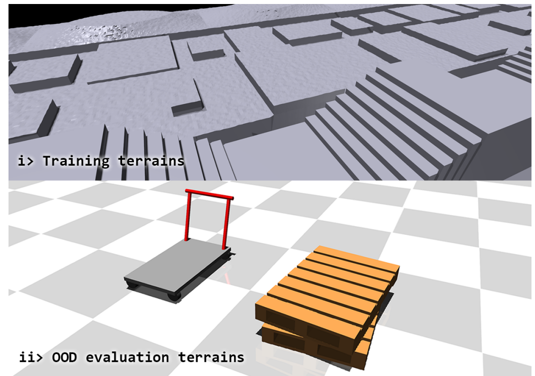

Fig. 3. Simulation terrains used in the experiments. The upper panel shows representative terrains used during training. The lower panel shows additional MuJoCo out-of-distribution terrains used for cross-simulator evaluation.

- **w/o GRF**: replaces gated residual fusion with a residual MLP block of matched depth and width.
- **w/o Highway Gate**: removes the highway output gate from recurrent fusion and decodes actions directly from the GRU feature.
- **HPL**: a perceptive locomotion baseline trained and evaluated under the same terrain protocol.

For architectural comparison, each policy is evaluated in 2000 parallel environments, and each rollout lasts up to 40 s. For stair foot-placement analysis, we use 2048 parallel stair rollouts. For highway-gate analysis, we log gate statistics from 4096 parallel environments under a fixed forward command. MuJoCo [31] is used only for an additional out-of-distribution evaluation in the architectural comparison. Fig. 3 shows the simulated training terrains together with the out-of-distribution evaluation terrains.

For the foothold-objective comparison, we replace the foothold placement reward in Eq. (23) with a foot contact quality reward (FCQR), following BeamDojo [10], while keeping the remaining training settings unchanged.

B. Simulation Experiments

This subsection reports the simulation results of CReF. We focus on architectural comparison, stair foothold distribution analysis, and highway-gate behavior analysis.

1) Architectural Comparison: Table II summarizes the architectural comparison. For the trained terrain categories, the stair entries are indexed by riser height and tread depth, while the gap and platform entries are indexed by gap width and platform height, all in centimeters. For stairs, gap, and platform, the OOD columns denote difficulty levels outside the training range. In the additional MuJoCo OOD evaluation, the values in parentheses denote the numbers of successful terrains out of 20 test terrains. The Overall score is computed from the stair, gap, and platform success rates only, excluding the additional MuJoCo OOD evaluation. Full CReF achieves the best overall terrain-traversal performance and consistently outperforms both the ablated variants and the perceptive baseline HPL. Its advantage becomes more

TABLE II
ARCHITECTURAL COMPARISON OF TERRAIN TRAVERSAL SUCCESS RATES (%).

Method	Stairs, SR				Gap, SR				Platform, SR				MuJoCo OOD	Overall (w/o MuJoCo)
	Easy (10/30 cm)	Medium (15/30 cm)	Hard (20/30 cm)	OOD (23/30 cm)	Easy (30 cm)	Medium (50 cm)	Hard (75 cm)	OOD (80 cm)	Easy (20 cm)	Medium (30 cm)	Hard (40 cm)	OOD (43 cm)		
Full CReF	**99.85**	**99.75**	**97.85**	**72.75**	**99.30**	**98.35**	**92.15**	**44.70**	**99.65**	**99.35**	**97.35**	**84.35**	100(*20*/20)	**90.45**
w/o Cross-Attn	96.25	87.50	78.70	34.00	97.10	97.70	83.00	8.60	99.40	98.75	96.90	64.80	95(*19*/20)	78.56
w/o GRF	99.60	93.40	83.60	49.10	98.00	98.25	90.05	29.95	99.45	99.15	96.20	68.55	95(*19*/20)	83.78
w/o Highway Gate	97.65	98.40	93.65	50.20	98.50	96.90	86.35	17.20	98.45	97.65	91.45	73.10	100(*20*/20)	83.29
HPL	96.30	81.45	71.05	27.40	97.45	94.40	78.25	20.85	98.05	92.65	81.80	55.15	5(*1*/20)	74.57

TABLE III
STAIR FAILURES IN ASCENT AND DESCENT.

Method	Easy		Medium		Hard		OOD	
	Up	Down	Up	Down	Up	Down	Up	Down
Full CReF	1	**2**	**2**	**3**	**5**	**49**	**90**	**454**
w/o Cross-Attn	**0**	75	47	203	88	338	496	824
w/o GRF	2	6	25	107	39	289	128	890
w/o Highway Gate	13	34	3	29	39	88	293	703
HPL	31	63	136	235	241	338	493	959

pronounced as terrain difficulty increases, especially on unseen and out-of-distribution settings, including the additional MuJoCo evaluation. These results indicate that the proposed framework improves both robustness and generalization.

Among the ablations, removing cross-modal attention causes the most significant degradation, particularly on hard and out-of-distribution terrains, showing that state-conditioned depth feature extraction is critical for perceptive traversal. Removing GRF or the highway gate also weakens performance, especially under more challenging conditions, indicating that both gated residual fusion (GRF) for multi-modal integration and recurrent fusion for state-dependent memory usage contribute to robust terrain locomotion.

Table III further separates stair failures into ascending and descending cases and reports the numbers of failed ascending and descending traversals, respectively, out of 2000 stair environments for each difficulty level. Across all methods, descending failures increase much more rapidly than ascending failures as stair difficulty rises, indicating that stair descent is the more challenging regime. Full CReF consistently yields the fewest descending failures, especially on hard and OOD stairs, which is consistent with improved anticipatory foothold selection and temporal integration.

2) Stair Foothold Distribution Analysis: Fig. 4 compares the stair touchdown distributions produced by the foothold placement reward and the FCQR baseline. The distributions are computed from converged rollouts on the hard stair setting, by pooling touchdown samples from both feet across all evaluated environments. The reported touchdown positions are the projected contact points of the foot-end geometric centers at touchdown. The proposed reward yields consistently tighter and more repeatable contact placement in both ascent and descent. On ascending stairs, it reduces the median absolute deviation from 3.0 cm to 1.5 cm and

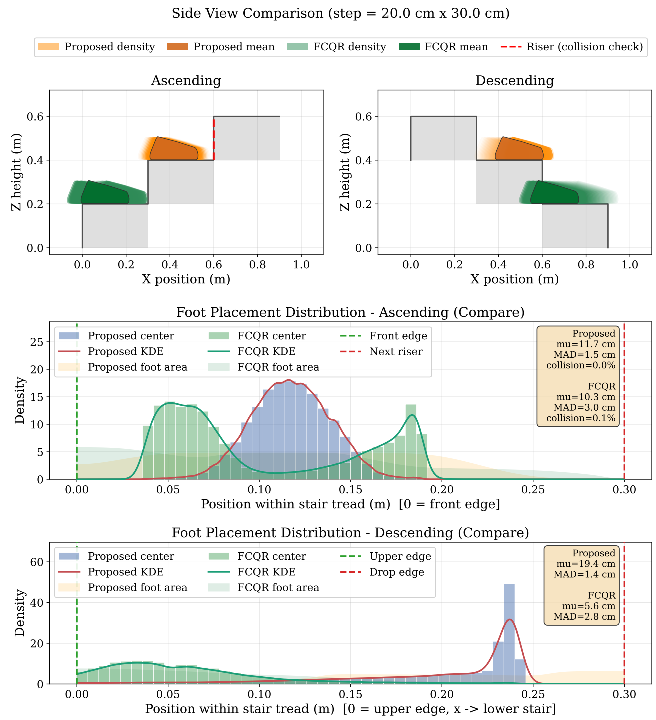

Fig. 4. Stair foothold distribution comparison between the proposed foothold placement reward and the FCQR baseline. The proposed reward produces more concentrated and repeatable touchdown distributions in both ascent and descent, reduces touchdown deviation within the stair tread, and eliminates the logged ankle-riser collisions in ascending rollouts.

eliminates the logged ankle-riser collisions. On descending stairs, it reduces the median absolute deviation from 2.8 cm to 1.4 cm.

The improvement is not merely a shift of the touchdown mean. Instead, the entire distribution becomes more concentrated, indicating that the reward improves contact precision and repeatability. This is consistent with its design: although computed at touchdown, it provides an anticipatory shaping objective toward locally supportable regions, rather than only discouraging unfavorable contacts after they occur.

3) Highway Gate Behavior Analysis: Fig. 6 analyzes the learned highway gate, where larger values indicate stronger reliance on recurrent features. For analysis, gate values are averaged over channels and then aggregated within each state group. We define *risky* states as timesteps satisfying $|\text{roll}| > 0.20$ rad or $|\text{pitch}| > 0.20$ rad, and treat the remaining timesteps as *stable* states. Flight phases are iden-

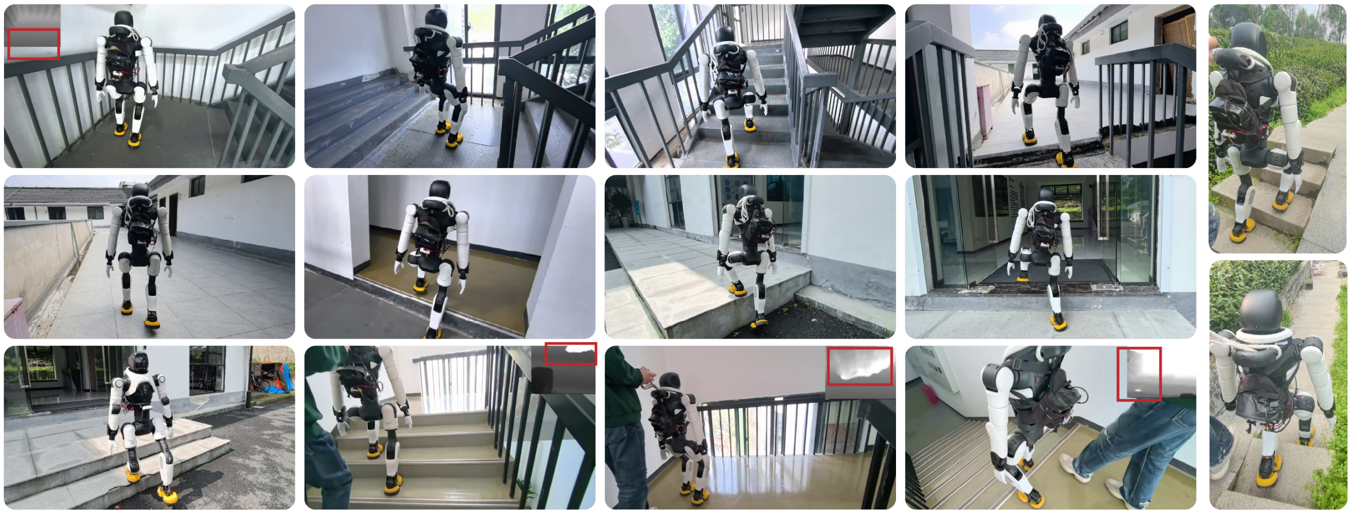

Fig. 5. Representative real-world rollouts of CReF across multiple terrains and deployment scenes. The figure includes stair traversal with side railings, entrance-step and platform-like transitions, outdoor pathways, and other real-world terrain configurations. Red boxes highlight examples where environmental factors introduce out-of-distribution depth observations, such as large invalid regions in the sensed depth image.

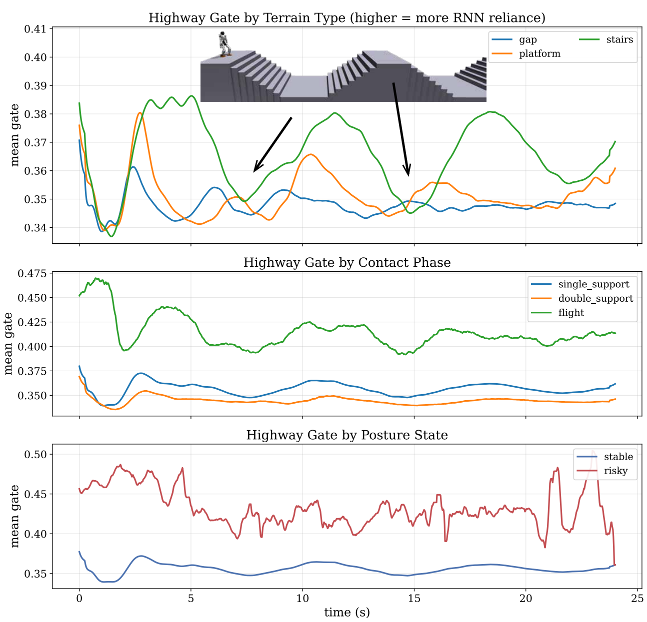

Fig. 6. Highway gate behavior under different traversal conditions. Higher gate values indicate stronger reliance on recurrent features.

tified from the foot-contact pattern. Across terrain types, the gate takes higher values on step-like terrains than on flat terrain, showing that the policy recruits temporal memory more strongly when precise terrain anticipation is required. Across contact phases, gate values are higher during flight than during stable support. Across posture states, risky states consistently induce larger gate activation than stable states.

These trends are consistent with the intended function of the highway gate: recurrent features are emphasized when instantaneous observations are less sufficient for reliable control, and attenuated when feedforward perception is adequate.

TABLE IV

INDOOR REAL-WORLD TRAVERSAL RESULTS.

Task	Setting	Success / Trials
Stairs	Indoor, 15 cm / 30 cm	20/20
Platform	Indoor, 40 cm	20/20
Gap	Indoor, 80 cm	18/20
OOD terrain	Indoor OOD scene	19/20

C. *Real-World Experiments*

Real-world experiments validate zero-shot transfer of CReF to a humanoid robot across stair traversal, platform ascent, gap crossing, and diverse indoor and outdoor scenes. Table IV reports representative indoor quantitative evaluations, where each task is assessed over 20 trials. For stairs, one trial consists of a complete ascent-descent traversal on indoor stairs with 15 cm rise and 30 cm tread. On the AGIBOT X2 Ultra, CReF achieves strong performance across these standardized indoor tasks, including successful traversal of a 40 cm platform and an 80 cm gap. Beyond these evaluations, CReF further demonstrates stable deployment in broader real-world environments. We refer to prior real-world humanoid results such as HPL only as context for task difficulty. In particular, HPL reports real-world perceptive parkour on the Unitree H1, including 0.42 m jump-up and 0.8 m leap tasks. However, we do not treat it as a direct like-for-like benchmark, as the robot platforms, hardware capabilities, and deployment setups differ.

Fig. 5 shows representative real-world rollouts. The policy stably climbs real stairs with side railings at a rise of 20 cm and a tread of 26 cm, even under changing illumination. It also remains effective in scenes with severe depth degradation, including large invalid depth holes caused by strong reflection, and in outdoor scenes containing dense vegetation, asymmetric boundaries, and other non-traversable structures that introduce substantial depth out-of-distribution effects. These results suggest that the learned representation preserves terrain-relevant cues while remaining robust to rail interference, reflective sensing failure, and cluttered real-world geometry.

IV. Conclusions

This paper presents CReF, which maps onboard proprioception and forward-facing depth directly to joint position targets without explicit geometric intermediates. By combining cross-modal attention, gated residual fusion, recurrent fusion, and a terrain-aware foothold placement reward, CReF improves terrain anticipation and foot placement consistency, with clear gains on stair traversal. Experiments in simulation and on hardware demonstrated strong terrain-traversal performance and zero-shot transfer to challenging real-world scenes.

At the same time, certain limitations still exist. Our system relies on active depth measurements from a RealSense sensor, which are sensitive to illumination and reflective surfaces, while depth alone does not preserve appearance and texture cues that may improve perceptual robustness. A natural direction for future work is therefore to investigate binocular RGB-based sensing, which can jointly provide depth structure and texture information for richer scene representations and stronger adaptability across diverse real-world environments.

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